

Complete OpenCV Topics List

1. Introduction & Setup

- What is OpenCV
- Installing OpenCV (pip install opencv-python)
- Reading, displaying, and saving images
- Working with video
- Image properties (shape, size, dtype, channels)

2. Basic Image Operations

- Image translation, rotation, resizing, flipping
- Cropping and ROI (Region of Interest)
- Color spaces (BGR, RGB, HSV, GRAY, etc.)
- Splitting and merging channels
- Drawing shapes (line, circle, rectangle, text)

3. Image Processing Techniques

- Filtering and blurring
- Thresholding (binary, adaptive, Otsu's method)
- Edge detection (Canny, Sobel, Laplacian)
- Morphological operations (erosion, dilation, opening, closing)
- Bitwise operations (AND, OR, XOR, NOT)

4. Geometric Transformations

- Affine and perspective transforms
- Image warping and remapping
- Rotation and scaling matrices

5. Contours & Shape Detection

- Finding and drawing contours
- Contour approximation
- Bounding rectangles and circles
- Convex hull and convexity defects
- Shape matching and moments

6. Image Segmentation

- Color-based segmentation (HSV masking)
- Watershed algorithm
- GrabCut segmentation
- Connected component analysis

7. Feature Detection & Description

- Corner detection (Harris, Shi-Tomasi)
- SIFT, SURF, ORB features
- Feature matching (BFMatcher, FLANN)

8. Video Analysis

- Motion detection
- Background subtraction
- Optical flow
- Object tracking (MeanShift, CamShift, MOSSE, CSRT)

9. Face & Object Detection

- Haar cascades (face, eyes, smile)
- HOG descriptor for people detection
- DNN module with YOLO, SSD, etc.

10. Machine Learning in OpenCV

- kNN, SVM, Decision Trees
- K-means clustering
- Handwritten digit recognition

11. Deep Learning Integration

- Loading pre-trained models (TensorFlow, Caffe, ONNX)
- Object detection (YOLO, SSD)
- Face recognition (OpenCV + face_recognition)

12. Utility & Advanced Topics

- Image histograms
- Template matching
- Image stitching (panorama)
- Camera calibration & distortion correction
- Augmented reality overlays

13. Projects & Applications

- Face & mask detection
- Gesture recognition
- Vehicle detection and counting
- Smart surveillance / motion alerts
- Virtual drawing board
- Attendance system using face recognition